DataCo Supply Chain Analytics

Sales, Profitability and Delivery Performance · January 2015 – January 2018

Executive Summary

Sales	Profit	Orders	AOV	On-time delivery
$36.04M	**$3.89M (10.8%)**	**64,385**	**$560**	**42.7%**

Cancellation
2.1%

Positives: stable ~$1.0M monthly sales through Sep-2017, consistent margins of 10-11% across markets and segments, a low 2.1% cancellation rate, and 57.2% repeat purchasing. **Risks:** only 42.7% of delivered lines are on time, First Class is late on every line, 33,784 order lines (18.7%) are loss-making, and sales drop sharply from Nov-2017 (likely data-capture related).

Priorities: 1) fix premium shipping schedules, 2) review discounts above 15%, 3) validate the late-2017 data before drawing conclusions.

Project Overview

Objective: identify drivers of sales, profit and delivery performance for DataCo's sports and outdoor supply chain. Data: DataCoSupplyChain_Cleaned.csv (180,519 order-item rows, 65,752 orders) and Data_Chain_Supply.sql (SQL Server). **No SQL result files were supplied**, so every figure here is calculated from the CSV with the SQL's business logic (pandas) and cross-checked against the in-browser dashboard. Tools: SQL Server, Python/pandas, HTML/CSS/JavaScript, Apache ECharts.

Data Preparation and Validation

- 180,519 rows, 0 duplicate rows, 3 null values (ZIP code). Dates 2015-01-01 to 2018-01-31.
- Grain: one row per order item; 45,902 orders have several items. Orders are counted distinct.
- **SQL correction:** the SQL treats Order Profit Per Order as possibly repeated per order. In this file it differs across items in all 45,902 multi-item orders and equals Benefit per order, so it is item-level and summed directly.
- Rules (kept from the SQL): sales, units, profit exclude CANCELED lines; suspected-fraud orders (4,062 lines) stay included; delivery rates use the three delivered labels (excluding

7,754 "Shipping canceled" lines); margin = total profit ÷ total sales.

- Label check: 0 mismatches between delivery label and Shipping Delay Days. Sales differs from Order Item Total + Discount on 1,224 rows (not investigated further).
- Delivery metrics include 93,658 lines on orders still pending, processing, on hold or in payment review.

KPI Analysis

Definitions are in kpi_dictionary.csv. Total orders (all statuses) 65,752; units sold 376,204; purchasing customers 20,464; average delay among late lines 1.62 days.

Sales and Revenue Findings

Peak month 2017-09 ($1.11M); lowest 2018-01 ($0.32M, a partial-data month). Sales fall -42% in Nov-2017 while orders stay ~2,000 and lines per order fall from 3.01 to 1.00, so this is most likely a data-capture change.

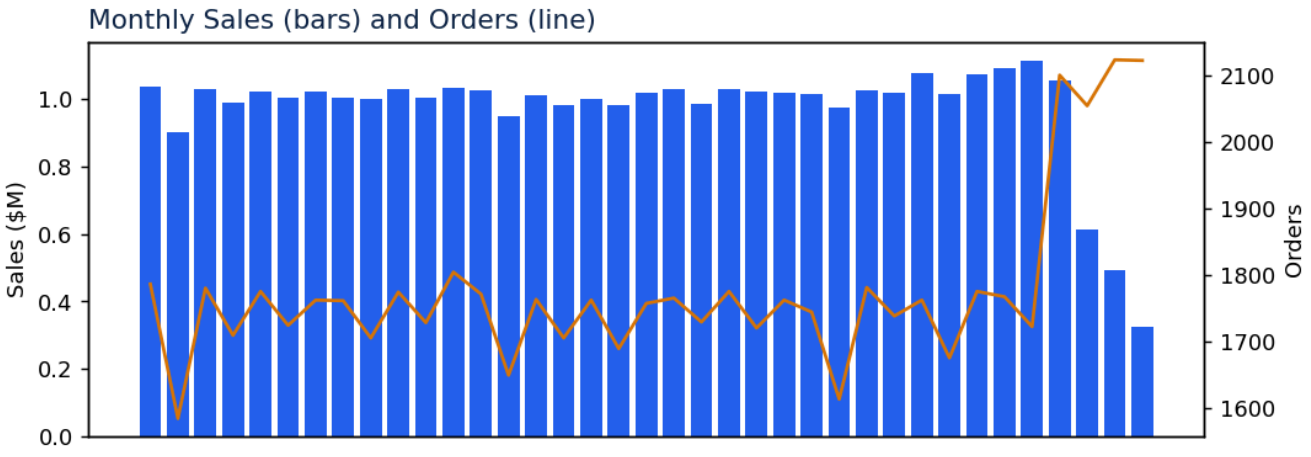
	Sales	Profit	Units	Orders	Margin %
Market					
Europe	10,657,721.2	1,155,766.3	103103	18189	10.8
LATAM	10,061,499.1	1,095,696.6	110500	16812	10.9
Pacific Asia	8,114,450.6	841,633.9	82086	17222	10.4
USCA	4,954,776.7	551,500.9	55414	8387	11.1
Africa	2,251,917.0	246,959.7	25101	3775	11.0

	Sales	Profit	Units	Orders	Margin %
Customer Segment					
Consumer	18,716,721.2	2,033,058.5	195280	33422	10.9
Corporate	10,937,747.7	1,178,170.6	114141	19435	10.8
Home Office	6,385,895.7	680,328.2	66783	11528	10.7

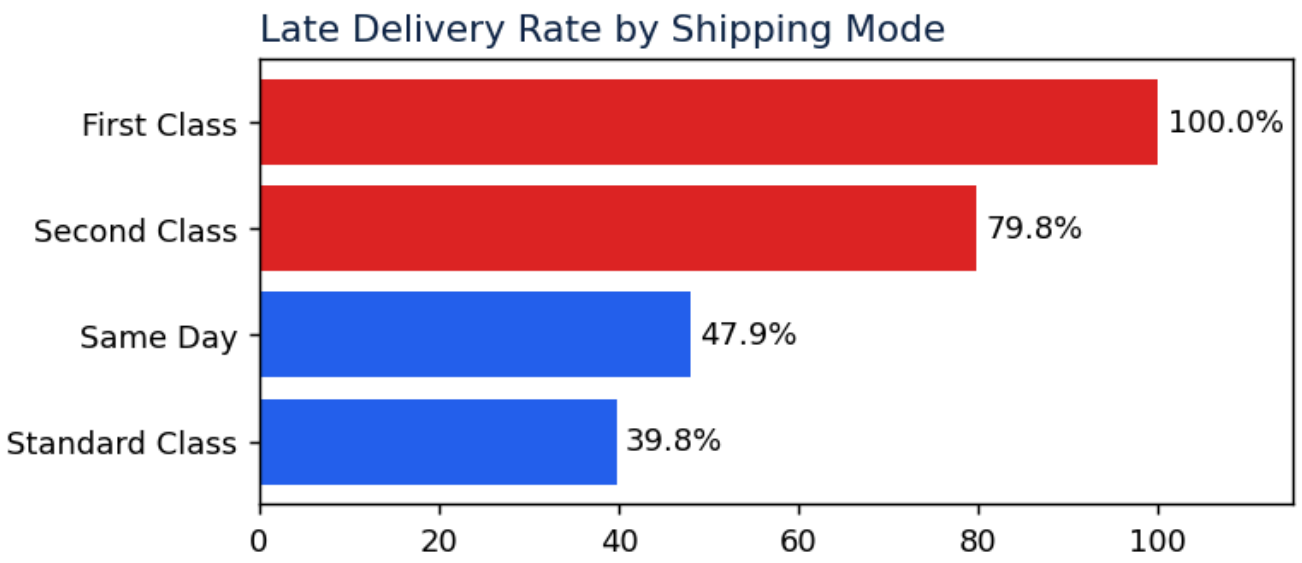
	Sales	Profit	Units	Orders	Margin %
Order Region					
Western Europe	5,777,446.1	621,095.9	55365	9813	10.8
Central America	5,559,567.4	602,710.2	60910	9212	10.8
South America	2,884,689.0	323,266.8	31758	4857	11.2
Northern Europe	2,108,327.4	229,334.7	19929	3634	10.9
Southern Europe	2,009,894.7	227,031.9	19163	3473	11.3
Oceania	1,980,029.8	199,228.2	20058	4275	10.1
Southeast Asia	1,889,479.4	207,352.6	18407	4257	11.0
Caribbean	1,617,242.7	169,719.6	17832	2743	10.5

Top categories:

	Sales	Profit	Units	Orders	Margin %
Category Name					
Fishing	6,795,260.4	744,205.1	16989	14874	11.0
Cleats	4,346,157.1	485,847.4	72304	19989	11.2
Camping & Hiking	4,038,330.9	416,031.3	13462	12057	10.3
Cardio Equipment	3,613,491.3	374,577.4	36758	11108	10.4
Women's Apparel	3,082,050.0	342,388.7	61641	17488	11.1
Water Sports	3,047,648.0	316,639.3	15209	13456	10.4
Men's Footwear	2,831,052.3	305,190.7	21779	18392	10.8
Indoor/Outdoor Games	2,830,867.2	311,681.9	56640	16295	11.0



Supply Chain Findings



Late rate by shipping mode (delivered lines):

Shipping Mode	Lines	Late %	Avg delay (late)
First Class	26,513.0	100.0	1.0
Second Class	33,806.0	79.8	2.5
Same Day	9,293.0	47.9	1.0
Standard Class	103,153.0	39.8	1.5

Regional late rates are narrow (51.6% to 60.1%), so shipping mode is a larger driver than region. Highest: Central Africa, Western Europe, South of USA .

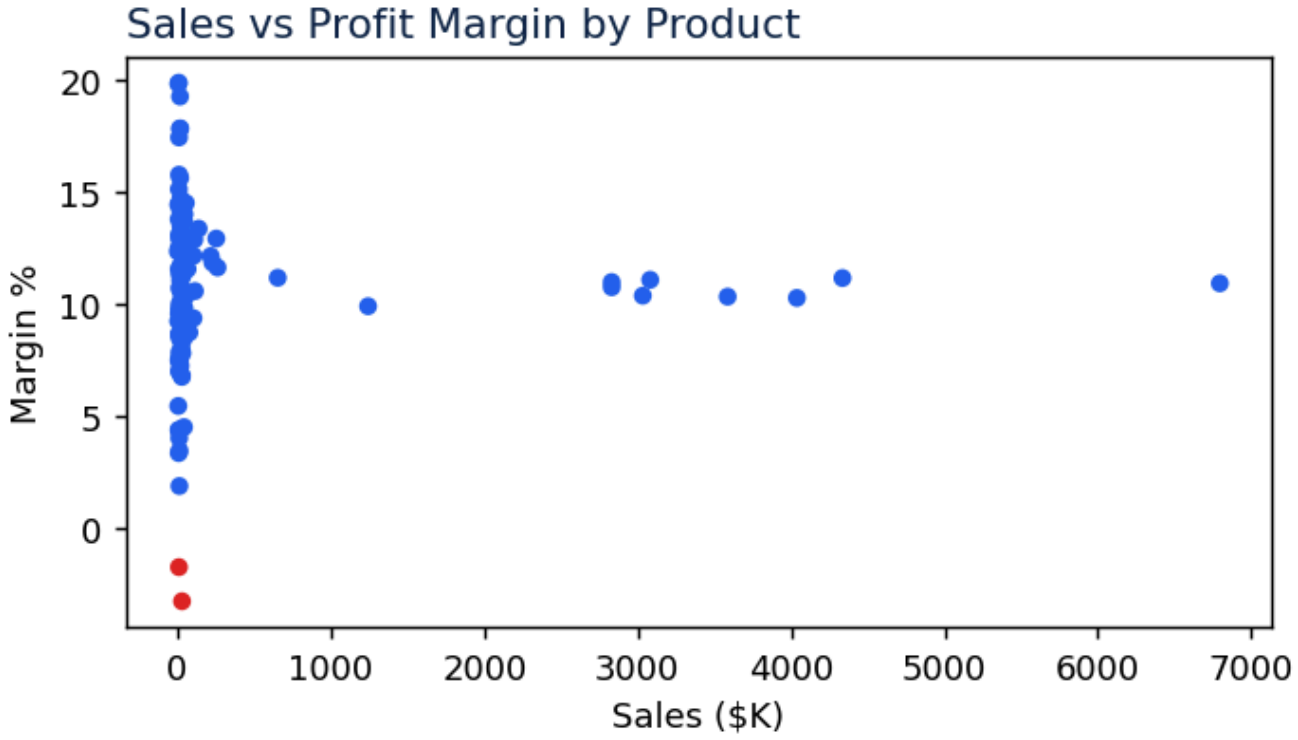
Product and Profitability Findings

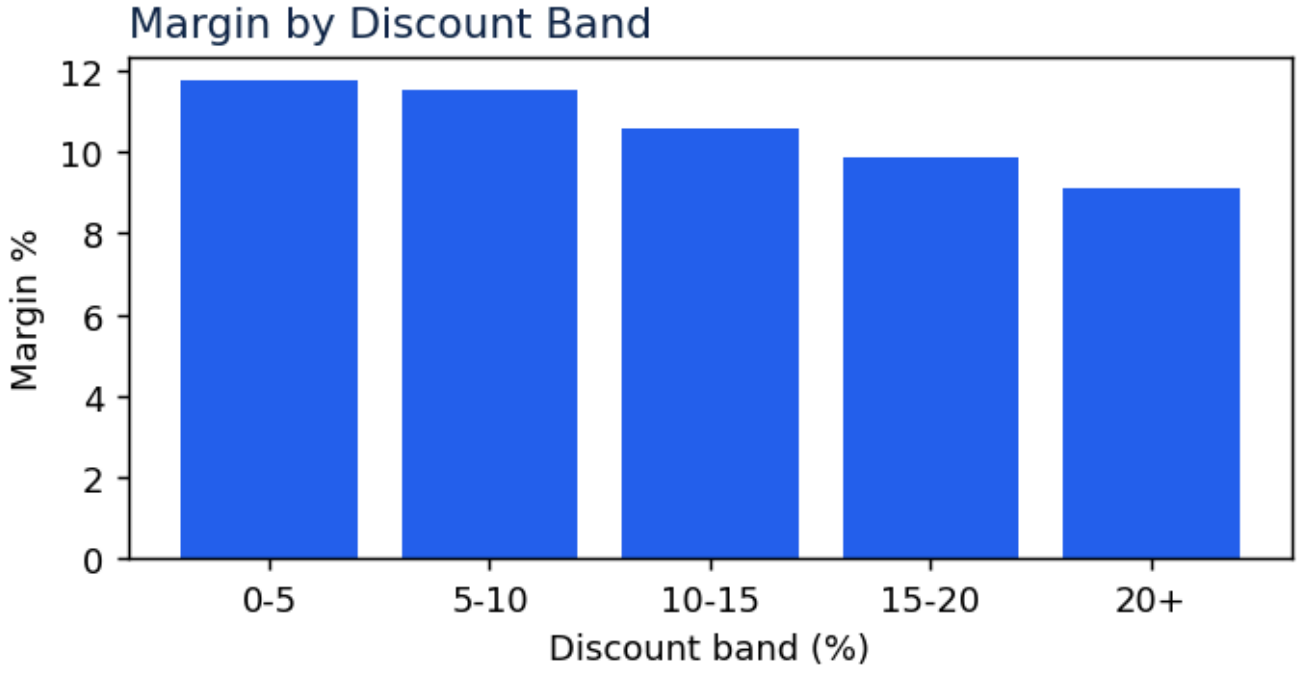
Lowest-profit products:

Product Name	Sales	Profit	Units	Orders	Margin %
SOLE E35 Elliptical	29,999.8	-965.1	15	15	-3.2
SOLE E25 Elliptical	9,999.9	-169.6	10	10	-1.7
GoPro HERO3+ Black Edition Camera	12,799.7	245.6	32	32	1.9
Diamondback Girls' Clarity 24 Hybrid Bike 201	8,399.7	284.4	28	27	3.4
Polar Loop Activity Tracker	5,717.4	313.7	52	51	5.5

Discount band vs margin (association only):

	Lines	Sales	Profit	Margin %
Order Item Discount Rate				
0-5%	49136	10,011,140.1	1,177,134.5	11.8
5-10%	39286	8,009,099.0	922,922.3	11.5
10-15%	29443	5,998,852.1	634,060.8	10.6
15-20%	39321	8,017,417.3	792,014.5	9.9
20%+	19641	4,003,856.2	365,425.2	9.1





Customer Findings

20,464 purchasing customers, average revenue $1,761, repeat rate 57.2%, top 10% of customers = 29.1% of revenue. Highest-revenue customer IDs: 791, 8766, 1657.

Business Recommendations

Fix First Class and Second Class schedules (Priority: High)
Evidence: First Class late rate 100.0%, Second Class 79.8% (Standard 39.8%); First Class is scheduled 1 day but takes 2.
Action: Review promised-day rules and carrier SLAs for premium modes
Expected benefit: Hypothesis: fewer late premium shipments.
KPI: Late rate by mode · Data needed: Carrier cost and delivery-date data

Review discount policy above 15%% (Priority: High)
Evidence: Margin falls from 11.8% (0-5%) to 9.1% (20%+) while average line sales stay flat (association, not causation).
Action: Test tighter discount caps on low-margin categories
Expected benefit: Hypothesis: margin recovery without volume loss.
KPI: Margin % by discount band · Data needed: Controlled discount test, product costs

Fix loss-making products and low-margin categories (Priority: Medium)
Evidence: 2 products have negative total profit (e.g. SOLE E35 Elliptical, SOLE E25 Elliptical); weakest category margins: Strength Training, Men's Clothing.
Action: Re-price or review sourcing and shipping cost
Expected benefit: Hypothesis: higher margin.
KPI: Product margin % · Data needed: Unit cost and return data

Resolve the late-2017 data-capture change (Priority: High)
Evidence: Sales fall 42% in Nov-2017 while orders stay ~2,000/month; lines per order drop from ~3.0 to ~1.0.
Action: Confirm with data owners before reporting a demand decline
Expected benefit: Avoids a wrong business conclusion.
KPI: Lines per order · Data needed: Source-system extract notes

Grow repeat purchasing (Priority: Medium)
Evidence: 57.2% of 20,464 customers ordered more than once; top 10% of customers = 29.1% of revenue.
Action: Loyalty/retention test for single-order customers
Expected benefit: Hypothesis: higher customer revenue.
KPI: Repeat purchase rate · Data needed: Campaign response data

Limitations and Next Steps

No cost, return, carrier or true delivery-date fields; no query result files supplied; delivery measures schedule adherence on order lines, not confirmed arrival. Findings show associations, not causes. Next: obtain unit costs, run a discount test, and confirm the late-2017 extract.

Conclusion

Revenue and margin are steady, but delivery reliability and discounting are the clearest levers, and late-2017 data needs validation before it informs decisions.